

ABOUT

- Passionate about learning new skills, writing clean code, and working with great people.
- Strong interest in designing high-performance, scalable, and architecturally solid software.
- Natural problem solver with an affinity towards complex technical challenges that require self-guidance.
- Meticulous about algorithm efficiency, optimization, correctness, safety, and SOLID principles.
- Drawn to powerful system programming languages such as C++ and Rust.

EDUCATION

BSc Games Software Engineering First Class Honours Bournemouth University

 09/2018 - 07/2021

- 3D Graphics Rendering and Computational Programming
- Deep Learning with Convolutional Neural Networks and Computer Vision
- Game Engine Design and Development
- 3D Mathematics and Physics for Computer Graphics

PROJECTS

LEARNING

[wgpu-hacking](#): Implementation of a white paper on rendering grass using instanced rendering in **Rust** and **wgpu**.

[wasm-game-of-life](#): Game of life implemented in **webassembly** and **Rust**.

[raytracing-weekend](#): Implementation of the famous book on raytracing in **Rust**.

[gpu-gems-scan](#): memory-coalesced parallel sum in **CUDA** with optimized bank conflicts for the Nvidia GPU architecture.

UNIVERSITY WORK

Hand Posture Recognition (dissertation): Convolutional NN (YOLO v4) that recognizes hand poses in varying degrees of self-occlusion and lighting in real time using a phone camera stream. Trained and deployed with **Python** and **TensorFlow**.

[eudaimonia](#): 3D Game Engine written in **C++** and **OpenGL** with a focus on performance and zero overhead abstraction. Data-driven and cache-coherent architecture. Physically based rendering.

EXPERIENCE

Graphics Software Engineer

Arm

 11/2024 - Present  United Kingdom

Semiconductor and software design company. Designs the Mali GPU chip.

- Research and development of graphics algorithms for the Arm Mali **GPU architecture**.
- Work on the end-to-end integration of hardware-accelerated ML algorithms such as Arm Neural Super Sampling (NSS) using **Vulkan** and **C++**.
- Developing an engine-agnostic SDK in **C++** with a **C** API that delivers graphics algorithms optimized for Arm Mali.
- Prototyping, evaluating and implementing graphics workloads in the area of AI, ray tracing, rendering, GPGPU, and others, within various game engines and frameworks (**Unreal Engine**, Vulkan Samples, in-house, **Unity**, etc.)
- Profiling and debugging of graphics and ML GPU workloads using Arm Performance Studio, RenderDoc and Nvidia Nsight.
- Shader programming with **HLSL** and **GLSL**.

Software Engineer

Hawk-Eye Innovations (Sony Sports)

 12/2021 - 11/2024  United Kingdom

Industry leader in sports visualization and real-time electronic officiating.

- Developed a Virtual Production system in **C++20** with graphics rendered in **Unreal Engine**, orchestrated via a **Qt6** GUI app, and a custom **TCP/IP** back-end architecture for communication with other parts of the software pipeline, such as video and tracking data ingestion and broadcast synchronization.
- Designed the architecture of the core application logic in **C++** from the ground up for the business's needs (real time data flow, broadcast synchronization, determinism, playback rate manipulation) using **SOLID** principles. This included crafting **design documents** and diagrams, introducing **architectural decision records**, presenting the idea to stakeholders, and writing all of the code for the architecture and base APIs.
- Ownership over design, implementation and maintenance of multiple features related to simulation logic, multi-threading, rendering, networking and operations.
- Personally led a transition to **Test-Driven Development** in the team with **GoogleTest** in our **C++** infrastructure and Unreal Test Automation for our rendering product. This was crucial to uphold sports integrity.
- Developed a **data driven** skeletal character animation system using a real time stream of tracking data (similar to motion capture, but fully proprietary).
- Peer reviewed code, on-boarded, and mentored teammates. Experienced cross-team dependency coordination and deliveries with deadlines for the largest sports events in the world, such as FIFA World Cup, Wimbledon, NBA, UEFA Champions League and many others.

Junior Developer

VISTA AR

 02/2020 - 08/2020  United Kingdom

A digital technology combining the real and virtual worlds in order to discover the invisible history of a site.

- Development of AR apps in **Unity** and **C#**.
- Profiling and performance optimization for mobile in high poly scenes.
- Shader programming with **HLSL** and Shader Graph.
- Performance analysis with **Python**.